Grade 11 – Term 1
C3 Practice Activity 3 with Solutions

Contents

Evaluated criterion

C3. Applies circular-arc probability density functions to calculate interval probabilities using the geometry of circles and circular segments.

1 Introduction

Problems:
1.1. Periodic variables and the cycle cut
1.2. Uniform and triangular pdf shapes
1.3. Nine-step geometric method

Criterion C2 constructed continuous pdfs and found probabilities from areas. Its uniform pdfs used rectangles, while its triangular pdfs used triangles and trapezoids. Criterion C3 applies exactly those geometric methods to variables that repeat. No new area method is needed; the new idea is to respect the cycle cut when an event passes from the end of one displayed cycle to the beginning of the next.

The word *circular* describes the **variable**: its values repeat after one period. The words *uniform* and *triangular* describe the **shape of its pdf**. A circular-uniform model therefore has a periodic variable and a rectangular pdf on one displayed cycle. A circular-triangular model has a periodic variable and straight sides that form one physical triangular peak around the cycle.

1.1 Periodic variables and the cycle cut

Compass direction repeats after one revolution, time of day repeats after one day, and rotation repeats when a device returns to its starting orientation. For example,

$$0^\circ \equiv 360^\circ, \qquad 0\ \text{h} \equiv 24\ \text{h}, \qquad f(x+P) = f(x),$$

where P is the period length. For radians, $P = 2\pi$; for degrees, $P = 360^\circ$; and for time of day in hours, $P = 24$ h.

Displaying one circle as an interval requires a *cut*. For example, $[0^\circ, 360^\circ)$ cuts the compass at north. The interval has a written beginning and end, but the physical compass has no endpoint. Thus 359° and 1° are only 2° apart across the cut. Likewise, 23:59 and 00:01 are close times even though their recorded numbers lie on opposite sides of the midnight cut.

The displayed interval is a recording convention, not a statement that the model stops beyond its endpoints. Consequently, do not automatically add an "otherwise" branch with zero density. Write the pdf on one chosen cycle and state its periodic extension. The equivalent endpoints describe the same physical value, and a wrapped event must be split only for area bookkeeping.

1.2 Uniform and triangular pdf shapes

The only area formulas required in this activity are

$$A_{\text{rectangle}} = bh, \qquad A_{\text{triangle}} = \frac{1}{2}bh, \qquad A_{\text{trapezoid}} = \frac{1}{2}(h_1 + h_2)b.$$

Here b is horizontal width and each h is a pdf height. A probability is the area of its shaded region, and the area over one complete cycle is 1.

For a uniform pdf on a period of length P, the whole cycle is a rectangle:

$$\text{Uniform: } 1 = Ph, \qquad h = \frac{1}{P}.$$

Equal-width arcs or time bands therefore have equal probabilities. This is a reasonable model when a controlled randomizer deliberately gives every part of a cycle the same chance, or when the model is explicitly introduced as a baseline for comparison. Periodicity alone does not imply uniformity.

For a triangular pdf on a period of length P, the complete physical peak has base P and height H:

$$\text{Triangular: } 1 = \frac{1}{2}PH, \qquad H = \frac{2}{P}.$$

If the mode lies inside the displayed interval, the graph rises to H and falls to zero at the cut. Straight-side heights can be found from line equations, similar triangles, or proportions, exactly as in C2.

If the physical peak lies at the cut, the unwrapped graph shows two triangular pieces: one descends from the peak at the left endpoint, and the other rises to the same peak at the equivalent right endpoint. If their bases are b_1 and b_2, then $b_1 + b_2 = P$ and

$$\underbrace{\frac{1}{2}b_1 H}_{\text{first displayed piece}} + \underbrace{\frac{1}{2}b_2 H}_{\text{second displayed piece}} = \frac{1}{2}PH = 1.$$

These are not two physical modes. They are the two displayed parts of one peak joined across the cut.

1.3 Nine-step geometric method

Use this routine for every problem.

1. Define the variable, units, period, and chosen cut.
2. Identify the rectangular or triangular shape.
3. Use total area 1 to find unknown heights.
4. Find intermediate heights with line equations, similar triangles, or proportions.
5. Write the pdf on one cycle and state its periodic extension.
6. Shade the requested region.
7. Split at the cycle cut or pdf breakpoints.
8. Add geometric areas, or subtract simple outside regions from 1.
9. Check $0 \le P(\text{event}) \le 1$ and interpret the result.

A circle has no intrinsic left or right tail. The context must name the direction and the cut. For example, "the clockwise sector from $300°$ to the end of the displayed cycle" means $[300°, 360°)$. A band within $30°$ of the cut is written as the wrapped event

$$[330°, 360°) \cup [0°, 30°].$$

The two written pieces are disjoint, so their labelled geometric areas are added.

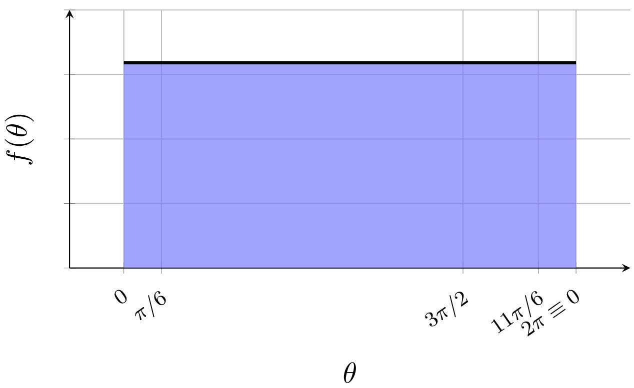
2 Circular Uniform Distributions

Problems:

2.1 Random sensor orientation

A university student in Bogotá is testing an augmented-reality app that places an object around the user. Let Θ be the object's clockwise orientation from the app's reference mark, measured in radians and recorded on $[0, 2\pi)$. Orientation is periodic because one full rotation returns to the same direction, so 0 and 2π are equivalent.

For each test, the app generates a direction by controlled randomization that gives equal chances to equal-width arcs. A circular-uniform pdf is therefore reasonable by design. The student checks the clockwise sector from $3\pi/2$ to the cut as one interface-test sector, and the wrapped band within $\pi/6$ of the reference mark as the app's alignment band.



Questions/Tasks.

C3.1 Construct the complete periodic pdf of Θ from the calibration narrative, using total rectangular area 1 to find its height and stating the period.

C3.2 Calculate geometrically the interface-test tail probability for the clockwise sector from $3\pi/2$ to the end of the displayed cycle.

C3.3 Calculate geometrically the alignment-band probability within $\pi/6$ of the reference mark, crossing the cycle cut.

C3.1 - Construct the periodic pdf.

The variable Θ is orientation in radians, its period is 2π, and its one-cycle support representation is $[0, 2\pi)$ with the cut at the reference mark. Controlled equal-arc randomization makes the pdf a rectangle of unknown height h. Its complete-cycle area gives

$$1 = A_{\text{rectangle}} = (2\pi)h, \qquad h = \frac{1}{2\pi}.$$

Thus, on the chosen cycle,

$$f_\Theta(\theta) = \frac{1}{2\pi}, \qquad 0 \le \theta < 2\pi,$$

and the complete periodic model is stated by

$$f_\Theta(\theta + 2\pi) = f_\Theta(\theta).$$

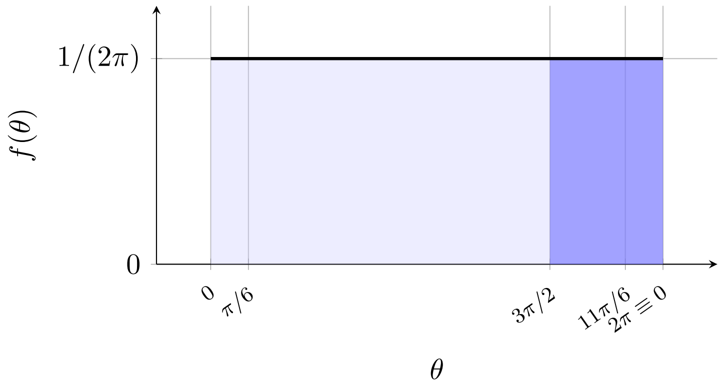
The period is 2π, and the geometric check is $(2\pi)(1/(2\pi)) = 1$.

C3.2 - Calculate the clockwise tail.

Relative to the cut at $0 \equiv 2\pi$, the interface-test tail means the clockwise sector $[3\pi/2, 2\pi)$. The shaded region below is one rectangle of width $2\pi - 3\pi/2 = \pi/2$ and height $1/(2\pi)$:

$$P\left(\frac{3\pi}{2} \le \Theta < 2\pi\right) = \underbrace{\left(2\pi - \frac{3\pi}{2}\right)\frac{1}{2\pi}}_{A_{\text{tail rectangle}}}$$
$$= \frac{1}{4} = 25\%.$$

The probability is valid because it lies between 0 and 1. One quarter of randomized tests are expected in the interface-test sector.



C3.3 - Calculate the wrapped alignment band.

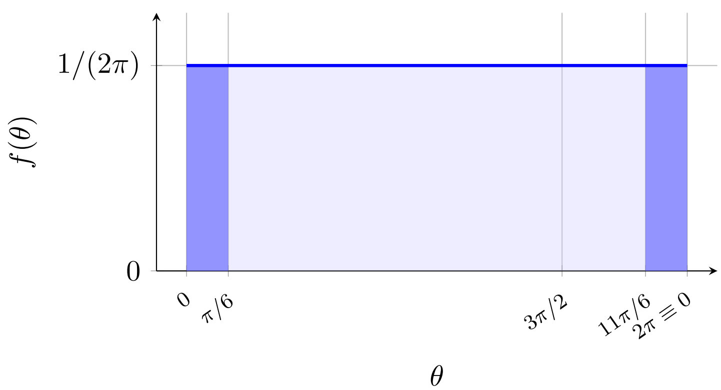
The band crosses the cut, so on $[0, 2\pi)$ it is

$$\left[\frac{11\pi}{6}, 2\pi\right) \cup \left[0, \frac{\pi}{6}\right].$$

The graph shades both written pieces. Each is a rectangle of width $\pi/6$:

$$\begin{aligned} P(\text{alignment band}) &= \underbrace{\frac{\pi}{6}\frac{1}{2\pi}}_{A_1=1/12} + \underbrace{\frac{\pi}{6}\frac{1}{2\pi}}_{A_2=1/12} \\ &= \frac{1}{6} \approx 16.7\%. \end{aligned}$$

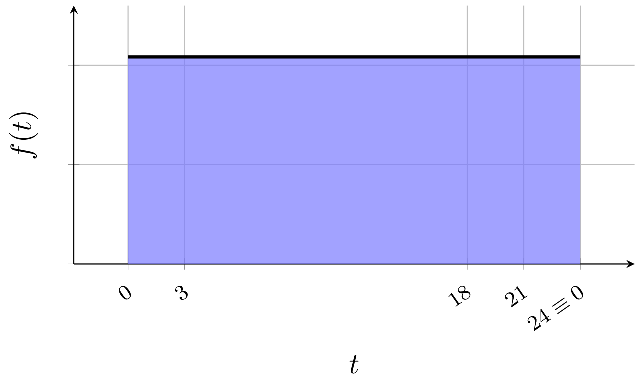
Thus about 16.7% of randomized orientations fall within $\pi/6$ of the reference direction, using both sides of the artificial cut.



2.2 Random daily system-check time

A student team in Bogotá maintains a digital platform for a university project and schedules one automated system check each day. Let T be the check's time of day in hours, recorded on $[0, 24)$ with the cut at midnight. Time of day is periodic because the 24-hour clock returns to the same time after one day: 0 h and 24 h are equivalent.

To avoid favoring any particular time window, the scheduler uses controlled randomization that gives equal chances to equal-length intervals. This makes a circular-uniform model reasonable. The team studies the probability that a check occurs from 18:00 to midnight and the probability that it occurs from 21:00 through 03:00, when a check would fall outside typical daytime work.



Questions/Tasks.

C3.1 Construct the complete periodic pdf of T from the scheduler narrative, using total rectangular area 1 to find its height and stating the period.

C3.2 Calculate geometrically the evening tail probability from 18:00 to the end of the displayed day.

C3.3 Calculate geometrically the overnight band probability from 21:00 through 03:00, crossing midnight.

C3.1 - Construct the periodic pdf.

The variable T is time of day in hours, its period is 24 h, and its one-cycle support representation is $[0, 24)$ with a midnight cut. Equal-window randomization gives a rectangular pdf of height h. Total area 1 gives

$$1 = A_{\text{rectangle}} = (24)h, \qquad h = \frac{1}{24}.$$

On the chosen recording cycle,

$$f_T(t) = \frac{1}{24}, \qquad 0 \le t < 24,$$

and its periodic extension is

$$f_T(t + 24) = f_T(t).$$

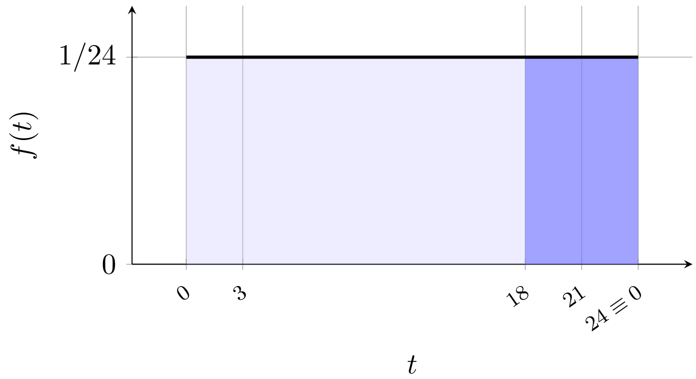
The period is 24 h and the rectangular area check is $24(1/24) = 1$.

C3.2 - Calculate the late-cycle tail.

The evening tail is defined relative to the midnight cut as $[18, 24)$, moving forward in clock time to the end of the displayed day. Its shaded shape is a width-6 rectangle:

$$P(18 \le T < 24) = \underbrace{(24 - 18)\frac{1}{24}}_{A_{\text{tail rectangle}}} = \frac{1}{4} = 25\%.$$

Therefore one quarter of randomized daily checks occur from 18:00 to midnight.



C3.3 - Calculate the overnight band.

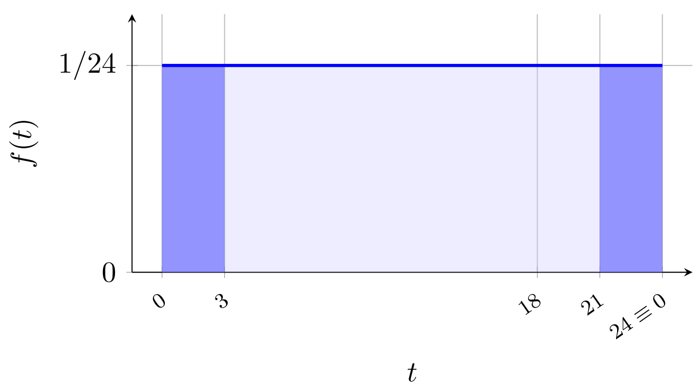
The time band crosses midnight, so the one-cycle representation is

$$[21, 24) \cup [0, 3].$$

The split occurs at the cycle cut. The shaded event consists of two width-3 rectangles:

$$\begin{aligned} P(\text{21:00 through 03:00}) &= \underbrace{(24-21)\frac{1}{24}}_{A_1=1/8} + \underbrace{(3-0)\frac{1}{24}}_{A_2=1/8} \\ &= \frac{1}{4} = 25\%. \end{aligned}$$

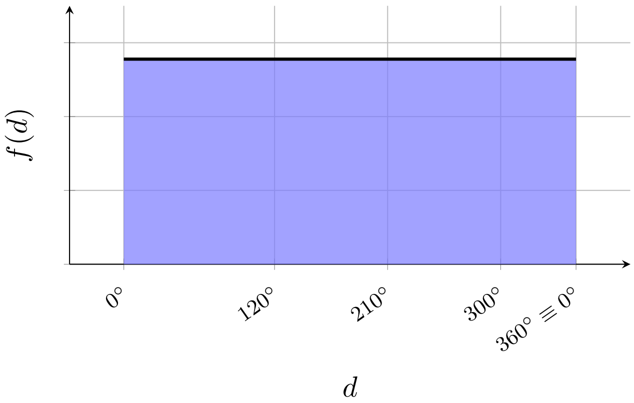
Thus 25% of the platform's randomized system checks occur in the 21:00–03:00 overnight band.



2.3 Randomized rotating-device direction

At a university event in Bogotá, a motorized camera is used to create 360° video clips. It records its direction D in degrees clockwise from its home mark on $[0°, 360°)$. Direction is periodic: after a 360° rotation the camera has returned to its home direction, so the displayed endpoints are equivalent.

For each clip, the controller deliberately randomizes the starting direction with no preferred sector, making a circular-uniform model appropriate by design. The clockwise sector from 300° to the cut is the side where the camera approaches its support cable, while the 120°–210° sector frames the event backdrop. The two probabilities help plan the camera setup.



Questions/Tasks.

C3.1 Construct the complete periodic pdf of D from the controller narrative, using total rectangular area 1 to find its height and stating the period.

C3.2 Calculate geometrically the support-cable tail probability for the clockwise sector from 300° to the end of the displayed cycle.

C3.3 Calculate geometrically the backdrop-viewing band probability for the target sector from 120° through 210°.

C3.1 - Construct the periodic pdf.

The variable D is direction in degrees, its period is 360°, and its one-cycle support representation is $[0°, 360°)$ with a cut at the home mark. Controlled equal-sector randomization gives a rectangular pdf. If its height is h, then

$$1 = A_{\text{rectangle}} = (360)h, \qquad h = \frac{1}{360}.$$

On one displayed cycle,

$$f_D(d) = \frac{1}{360}, \qquad 0 \le d < 360,$$

with density measured per degree. The periodic extension is

$$f_D(d + 360°) = f_D(d).$$

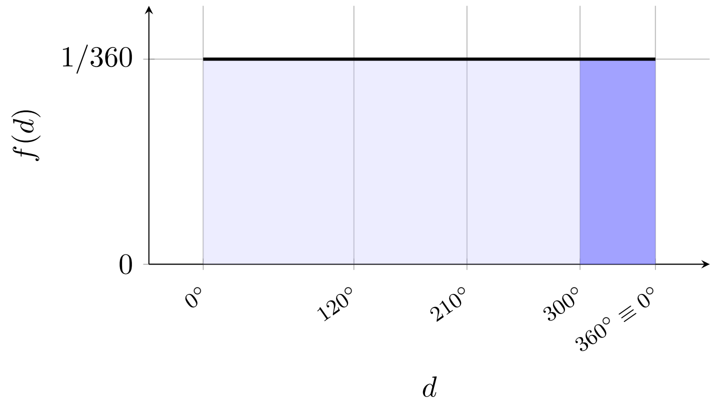
The period is 360°, and $360(1/360) = 1$ verifies the total area.

C3.2 - Calculate the directional tail.

The support-cable tail is not an intrinsic right tail of a circle. In this context it is specifically the clockwise sector $[300°, 360°)$ from 300° to the home-mark cut. The shaded rectangle has width 60°:

$$P(300° \le D < 360°) = \underbrace{(360 - 300)\frac{1}{360}}_{A_{\text{tail rectangle}}} = \frac{1}{6} \approx 16.7\%.$$

About 16.7% of randomized camera directions fall in the support-cable sector.

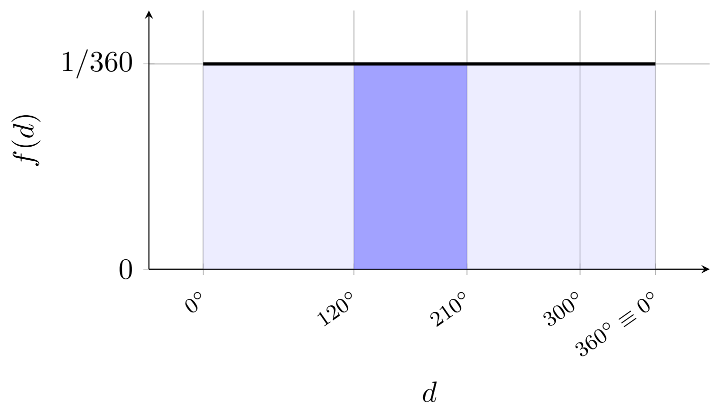


C3.3 - Calculate the target sector.

The target band $[120°, 210°]$ does not cross the cut or a pdf breakpoint. Its shaded region is one width-90° rectangle:

$$P(120° \leq D \leq 210°) = \underbrace{(210 - 120)\frac{1}{360}}_{A_{\text{target rectangle}}} = \frac{1}{4} = 25\%.$$

The randomized camera frames the backdrop sector in one quarter of inspections, which gives the student team a baseline for expected backdrop coverage.



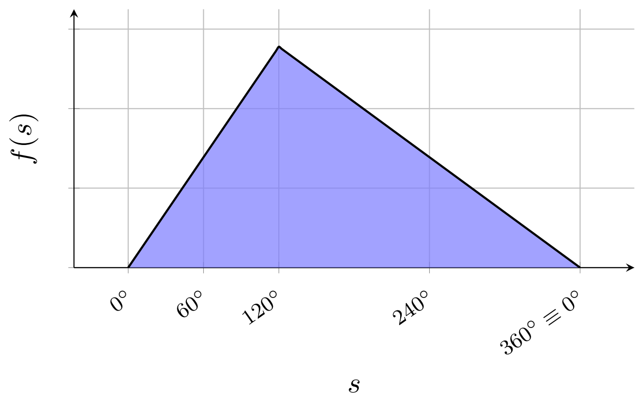
3 Circular Triangular Distributions

Problems:

3.1 Scanner direction with an interior mode

In a technical-education robotics lab in Bogotá, a motorized scanner records its clockwise direction S in degrees on $[0°, 360°)$, with the cut at north. Direction is periodic because 0° and 360° are the same physical direction.

For a practice scan, the programmed profile makes north least likely, increases the direction density at a constant rate to a preferred direction of 120°, and then decreases it at a constant rate back to zero at the cut. A circular-triangular pdf is reasonable because the programmed profile has one preferred direction and straight changes on both sides. The sector from 240° to the cut faces a reflective window, while the 60°–240° band covers the project-display area being scanned.



Questions/Tasks.

C3.1 Construct the complete periodic pdf of S from the scan profile, using total triangular area 1 to find all heights and stating the period.

C3.2 Calculate geometrically the reflective-window tail probability for the clockwise sector from 240° to the end of the displayed cycle.

C3.3 Calculate geometrically the project-display band probability from 60° through 240°.

C3.1 - Construct the periodic pdf.

The variable S is direction in degrees, the period is 360°, and its one-cycle support representation is $[0°, 360°)$ with its cut at north. The graph is a triangle with mode 120°, zero height at the cut, and unknown peak H. Total area 1 gives

$$1 = A_{\text{triangle}} = \frac{1}{2}(360)H, \qquad H = \frac{1}{180}.$$

The rising line through $(0, 0)$ and $(120, 1/180)$ has slope

$$\frac{1/180 - 0}{120 - 0} = \frac{1}{21600},$$

so its equation is $s/21600$. The falling line through $(120, 1/180)$ and $(360, 0)$ has slope $-1/43200$; using $(360, 0)$ gives $(360 - s)/43200$. Therefore, on one cycle,

$$f_S(s) = \begin{cases} \dfrac{s}{21600}, & 0 \le s \le 120, \\ \dfrac{360 - s}{43200}, & 120 < s < 360. \end{cases}$$

Its periodic extension is

$$f_S(s + 360°) = f_S(s).$$

The areas verify the model geometrically:

$$\underbrace{\frac{1}{2}(120)\frac{1}{180}}_{1/3} + \underbrace{\frac{1}{2}(240)\frac{1}{180}}_{2/3} = 1.$$

C3.2 - Calculate the reflective-window tail.

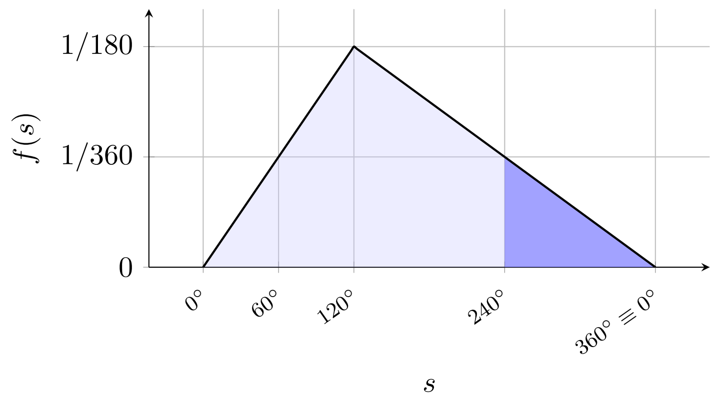
Relative to the north cut, this tail is the clockwise sector $[240°, 360°)$. The line equation gives

$$f_S(240) = \frac{360 - 240}{43200} = \frac{1}{360}.$$

The shaded tail is a triangle ending at zero height at the cut, with base 120° and height 1/360:

$$P(240° \le S < 360°) = \underbrace{\frac{1}{2}(360 - 240)\frac{1}{360}}_{A_{\text{tail triangle}}} = \frac{1}{6} \approx 16.7\%.$$

Thus the scanner points toward the reflective-window sector on about 16.7% of programmed scans.



C3.3 - Calculate the project-display band.

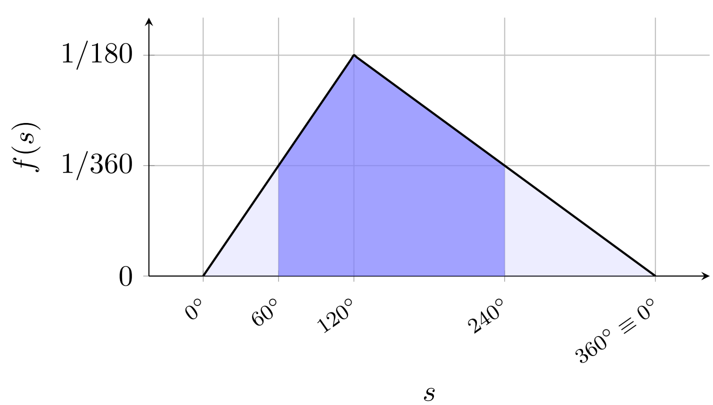
The band $[60°, 240°]$ crosses the pdf breakpoint at the mode, so it is split at 120°. The boundary heights are

$$f_S(60) = \frac{1}{360}, \qquad f_S(120) = \frac{1}{180}, \qquad f_S(240) = \frac{1}{360}.$$

The two shaded pieces are trapezoids:

$$\begin{aligned} P(60° \le S \le 240°) &= \underbrace{\frac{1}{2}\left(\frac{1}{360} + \frac{1}{180}\right)(60)}_{A_1 = 1/4} \\ &\quad + \underbrace{\frac{1}{2}\left(\frac{1}{180} + \frac{1}{360}\right)(120)}_{A_2 = 1/2} \\ &= \frac{3}{4} = 75\%. \end{aligned}$$

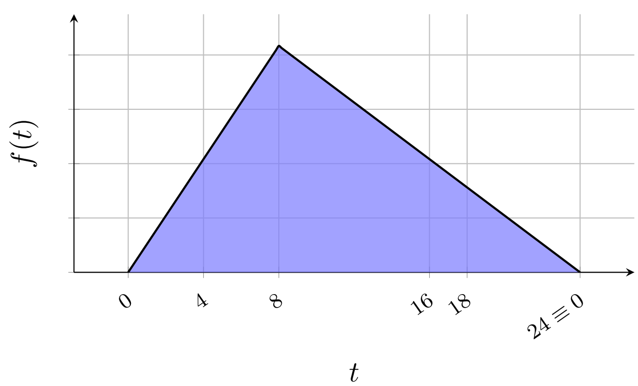
The project-display band covers three quarters of the programmed scan directions.



3.2 Daily peak-activity time

A university study platform records T, the time of day at which its daily student activity reaches its peak, in hours on $[0, 24)$. Time of day is periodic because the clock repeats every 24 hours and midnight is both 0 h and 24 h. The model uses midnight as the cut and as the least likely peak-activity time.

For this planning model, the density rises at a constant rate from zero at midnight to a mode at 08:00, then falls at a constant rate to zero at the next midnight. A circular-triangular pdf represents this single daily peak and the specified straight changes. The platform team studies peaks after 18:00 and the 04:00–16:00 band when planning when student activity is most likely to reach its daily maximum.



Questions/Tasks.

C3.1 Construct the complete periodic pdf of T from the activity narrative, using total triangular area 1 to find all heights and stating the period.

C3.2 Calculate geometrically the late-activity tail probability from 18:00 to the end of the displayed day.

C3.3 Calculate geometrically the daytime-activity band probability from 04:00 through 16:00.

C3.1 - Construct the periodic pdf.

The variable T is daily peak-activity time in hours. Its period is 24 h, and its one-cycle support representation is $[0, 24)$ with the cut at the least likely time, midnight. The pdf is a triangle with mode 8 and unknown peak H. Its complete-cycle area gives

$$1 = A_{\text{triangle}} = \frac{1}{2}(24)H, \qquad H = \frac{1}{12}.$$

The line from $(0, 0)$ to $(8, 1/12)$ has slope $1/96$, so the rising equation is $t/96$. The line from $(8, 1/12)$ to $(24, 0)$ has slope $-1/192$; through $(24, 0)$ its equation is $(24 - t)/192$. On one cycle,

$$f_T(t) = \begin{cases} \dfrac{t}{96}, & 0 \le t \le 8, \\ \dfrac{24 - t}{192}, & 8 < t < 24. \end{cases}$$

The periodic extension is

$$f_T(t + 24) = f_T(t).$$

The two triangular pieces have total area

$$\underbrace{\frac{1}{2}(8)\frac{1}{12}}_{1/3} + \underbrace{\frac{1}{2}(16)\frac{1}{12}}_{2/3} = 1.$$

C3.2 - Calculate the late-activity tail.

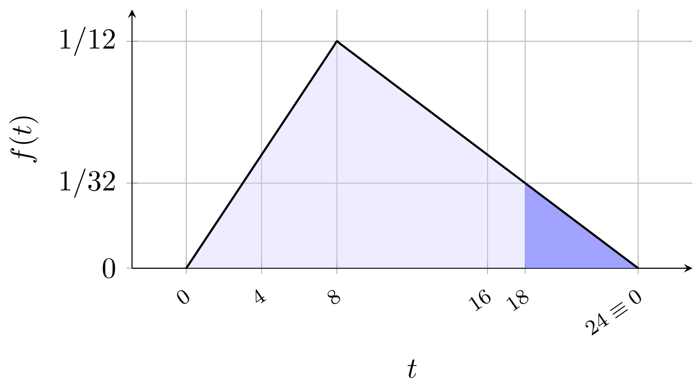
The late-cycle tail is defined as $[18, 24)$, moving forward in clock time from 18:00 to the midnight cut. Its starting height is

$$f_T(18) = \frac{24 - 18}{192} = \frac{1}{32}.$$

The shaded tail is a triangle of base 6 h and height 1/32:

$$P(18 \le T < 24) = \underbrace{\frac{1}{2}(24 - 18)\frac{1}{32}}_{A_{\text{tail triangle}}} = \frac{3}{32} = 9.375\%.$$

Therefore about 9.4% of days have a modeled peak late enough to place the daily activity peak after 18:00.



C3.3 - Calculate the daytime-activity band.

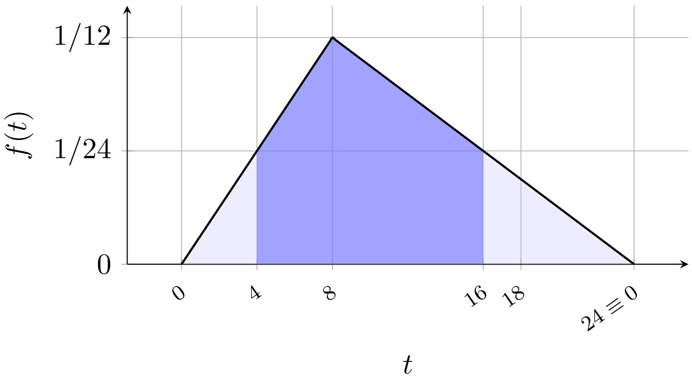
The band $[4, 16]$ crosses the mode at 8, which is a pdf breakpoint. Its three required heights are

$$f_T(4) = \frac{1}{24}, \qquad f_T(8) = \frac{1}{12}, \qquad f_T(16) = \frac{1}{24}.$$

Splitting the shaded band at 8 produces two trapezoids:

$$\begin{aligned} P(4 \le T \le 16) &= \underbrace{\frac{1}{2}\left(\frac{1}{24} + \frac{1}{12}\right)(8 - 4)}_{A_1 = 1/4} \\ &\quad + \underbrace{\frac{1}{2}\left(\frac{1}{12} + \frac{1}{24}\right)(16 - 8)}_{A_2 = 1/2} \\ &= \frac{3}{4} = 75\%. \end{aligned}$$

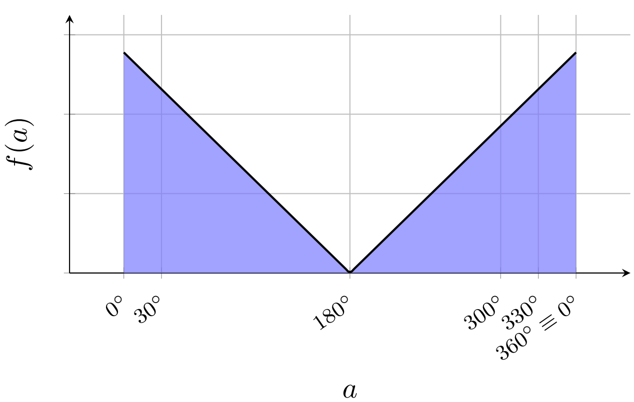
Thus the 04:00–16:00 band contains the daily activity peak on 75% of modeled days.



3.3 Direction with its peak at the cut

At a live-music event, a programmable stage light records its direction A in degrees clockwise from north on $[0°, 360°)$. Direction is periodic, so north is both $0°$ and the equivalent endpoint $360°$; the physical direction does not stop at the recording cut.

The lighting program most strongly favors north, reduces the direction density at a constant rate to zero at south $(180°)$, and then increases it at the same constant rate back toward north. A circular-triangular pdf is reasonable for this symmetric programmed preference. The sector from $300°$ to the cut covers one side of the audience area, while the band within $30°$ of north is a narrow stage-target zone.



Questions/Tasks.

C3.1 Construct the complete periodic pdf of A, showing how two displayed triangular pieces form one physical peak at the cut, using their combined area 1 to find all heights, and stating the period.

C3.2 Calculate geometrically the audience-side tail probability for the clockwise upper-end sector from $300°$ to the end of the displayed cycle.

C3.3 Calculate geometrically the stage light's wrapped target-band probability on $[330°, 360°) \cup [0°, 30°]$.

C3.1 - Construct the periodic pdf.

The variable A is stage-light direction in degrees, its period is $360°$, and its one-cycle support representation is $[0°, 360°)$ with the cut at the physical mode, north. The unwrapped graph shows a descending triangle on $[0°, 180°]$ and an ascending triangle on $[180°, 360°)$. Their endpoint heights both represent the same northward peak H.

The combined area is 1:

$$1 = \underbrace{\frac{1}{2}(180)H}_{\text{left displayed triangle}} + \underbrace{\frac{1}{2}(180)H}_{\text{right displayed triangle}} = 180H, \qquad H = \frac{1}{180}.$$

The first side has slope $-H/180 = -1/32400$ and passes through $(180, 0)$, giving $(180 - a)/32400$. The second has slope $1/32400$ and also passes through $(180, 0)$, giving $(a - 180)/32400$. On one cycle,

$$f_A(a) = \begin{cases} \dfrac{180 - a}{32400}, & 0 \le a \le 180, \\ \dfrac{a - 180}{32400}, & 180 < a < 360. \end{cases}$$

The complete model repeats according to

$$f_A(a + 360^\circ) = f_A(a).$$

Each displayed triangle has area $1/2$, so their areas sum to 1. The peak at 0° and the height approached at 360° are one physical peak, not two modes.

C3.2 - Calculate the upper-end directional tail.

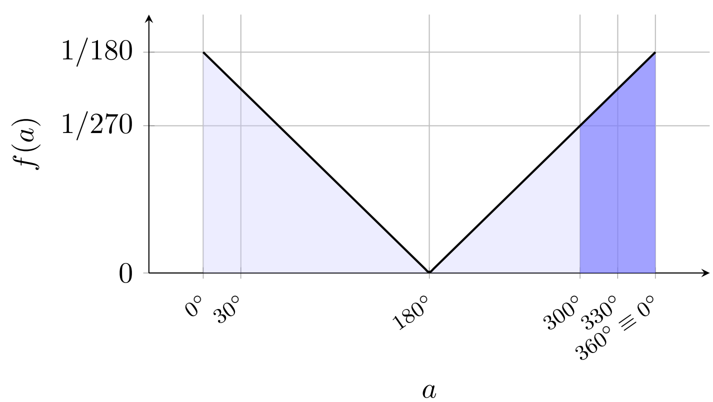
Relative to the north cut, the audience-side tail is the clockwise sector $[300^\circ, 360^\circ)$. Its parallel heights are

$$f_A(300) = \frac{120}{32400} = \frac{1}{270}, \qquad f_A(360^-) = \frac{1}{180}.$$

The upper endpoint is excluded from the written cycle, but its approached height completes the geometric edge at the equivalent north direction. The shaded tail is a trapezoid of width 60°:

$$P(300^\circ \le A < 360^\circ) = \underbrace{\frac{1}{2}\left(\frac{1}{270} + \frac{1}{180}\right)(60)}_{A_{\text{tail trapezoid}}}$$
$$= \frac{5}{18} \approx 27.8\%.$$

Thus about 27.8% of programmed directions use the audience-side sector.



C3.3 - Calculate the wrapped target band.

The physical target cone is continuous around north, but the chosen recording cycle splits it at the cut:

$$[330°, 360°) \cup [0°, 30°].$$

The outer boundary heights are equal:

$$f_A(330) = \frac{150}{32400} = \frac{1}{216}, \qquad f_A(30) = \frac{150}{32400} = \frac{1}{216}.$$

Each shaded piece is a trapezoid of width 30°, with heights 1/216 and 1/180 at its parallel sides:

$$\begin{aligned} P(\text{target cone}) &= \underbrace{\frac{1}{2}\left(\frac{1}{216} + \frac{1}{180}\right)(30)}_{A_1 = 11/72} \\ &\quad + \underbrace{\frac{1}{2}\left(\frac{1}{180} + \frac{1}{216}\right)(30)}_{A_2 = 11/72} \\ &= \frac{11}{36} \approx 30.6\%. \end{aligned}$$

The result lies between 0 and 1. About 30.6% of programmed stage-light directions fall within 30° of north, using both sides of the cut.